



**Department of Electrical and Computer Engineering
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Junior Design Project

Gamified To-do List

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ABSTRACT

Gamified To-Do List

A gamified to-do list is a task management system designed to enhance productivity by incorporating game elements such as rewards, and achievements. By adding game-like elements we aim to boost the engagement necessary to sustain consistent use, leading to increased motivation and productivity. The development process involved analyzing existing gamified systems, identifying user needs, and integrating features like virtual rewards, and competitive leaderboards. The integration of gamification principles encouraged sustained user engagement and made routine tasks more enjoyable.

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Problem Statement

Traditional to-do lists are effective for organizing tasks but often have low user retention. Many users struggle with procrastination and lack of accountability. This results in incomplete tasks, reduced productivity, and difficulty maintaining consistent usage of such tools.

Existing solutions lack features that actively incentivize users to complete their tasks in a rewarding or engaging manner. Traditional task management systems lose effectiveness by not being able to motivate the users.

There is a need for a solution that not only organizes tasks but also motivates users to consistently engage with the system and achieve their goals. A gamified to-do list addresses this problem by integrating game-like mechanics to enhance user experience, sustain engagement, and improve task completion rates.

Chapter 1 Introduction

1.1 Background motivation

Task management is a critical aspect of productivity, helping individuals organize, prioritize, and execute their responsibilities efficiently. Conventional to-do lists have long been a popular tool for this purpose, but they often overlook the psychological factors that influence task completion, such as motivation, procrastination, and user engagement. Lack of motivation is a significant barrier to productivity, with individuals often abandoning traditional task management systems.

One promising approach to tackle these issues is gamification. The application of game-design elements in non-game contexts. By incorporating elements such as points, achievements, leaderboards, and rewards, gamification leverages intrinsic and extrinsic motivation to encourage user participation and sustained engagement.

There is a need for effective task management tools, especially in modern lifestyles characterized by high workloads with competing priorities and distractions. A gamified to-do list seeks to bridge the gap between utility and motivation, transforming task management into an engaging and rewarding experience. This project's motivation is the potential of gamification to address the shortcomings of traditional systems and its ability to make productivity tools enjoyable and habit-forming. By fostering consistency and enhancing the sense of accomplishment, a gamified approach can significantly improve users' ability to manage their tasks and achieve their goals.

1.2 Purpose of Goal of the project

The goal of this project is to develop a gamified to-do list application that retains user engagement. By creating a system that not only organizes tasks efficiently but also motivates users to complete them by incorporating gamification elements such as rewards, progress tracking, and competitive features we hope to keep users engaged and stop frequent tool switching.

1.3 Organization of the report

This report is organized into seven chapters. The first chapter introduces the project and its motivation. Chapter 2 reviews existing case studies on gamified systems and their limitations. Chapter 3 outlines the methodology. Chapter 4 discusses the results, analysis, and findings. Chapter 5 explores the societal, environmental, and sustainability impacts of the project. Chapter 6 addresses project management, and Chapter 7 concludes the report.

Chapter 2 Research Literature Review

2.1 Existing Research and Limitations

Gamification is the use of game elements, and game design principles, including rewards, achievements, and competition in contexts other than games. Gamified task management makes users take actions that increase their level of engagement, for example, beating procrastination and lack of motivation.

Case Study: Forest

Forest is a focus and productivity application that makes use of game principles to assist users in trying to concentrate on their target tasks. Users will virtually plant a tree that will eventually grow as long as they don't leave the app and wither if they exit too soon. Users eventually would be able to grow a virtual forest with hows how productive they are in doing those habits.

Limitations:

- This program is designed to be focused rather than able to fully manage every task.
- Users do not have the ability to attach task details as well as a task list and a feature for prioritizing such tasks.

Case Study: Todoist Karma

Todoist Karma is an inbuilt feature of the award-winning Todoist app that enables users to earn points for completing a range of tasks increasing their productivity.

Limitations:

- As it has no competitive, or collaborative features, it appears to have no value for users who wish to be more socially engaged.

Relevance to the Project

This project is designed to fill such voids that have been noticed in the aforementioned systems by creating a to-do list that is gamified and combines the effective features of both Forest and Todoist Karma and other such features such as.

Chapter 3 Methodology

3.1 System Design

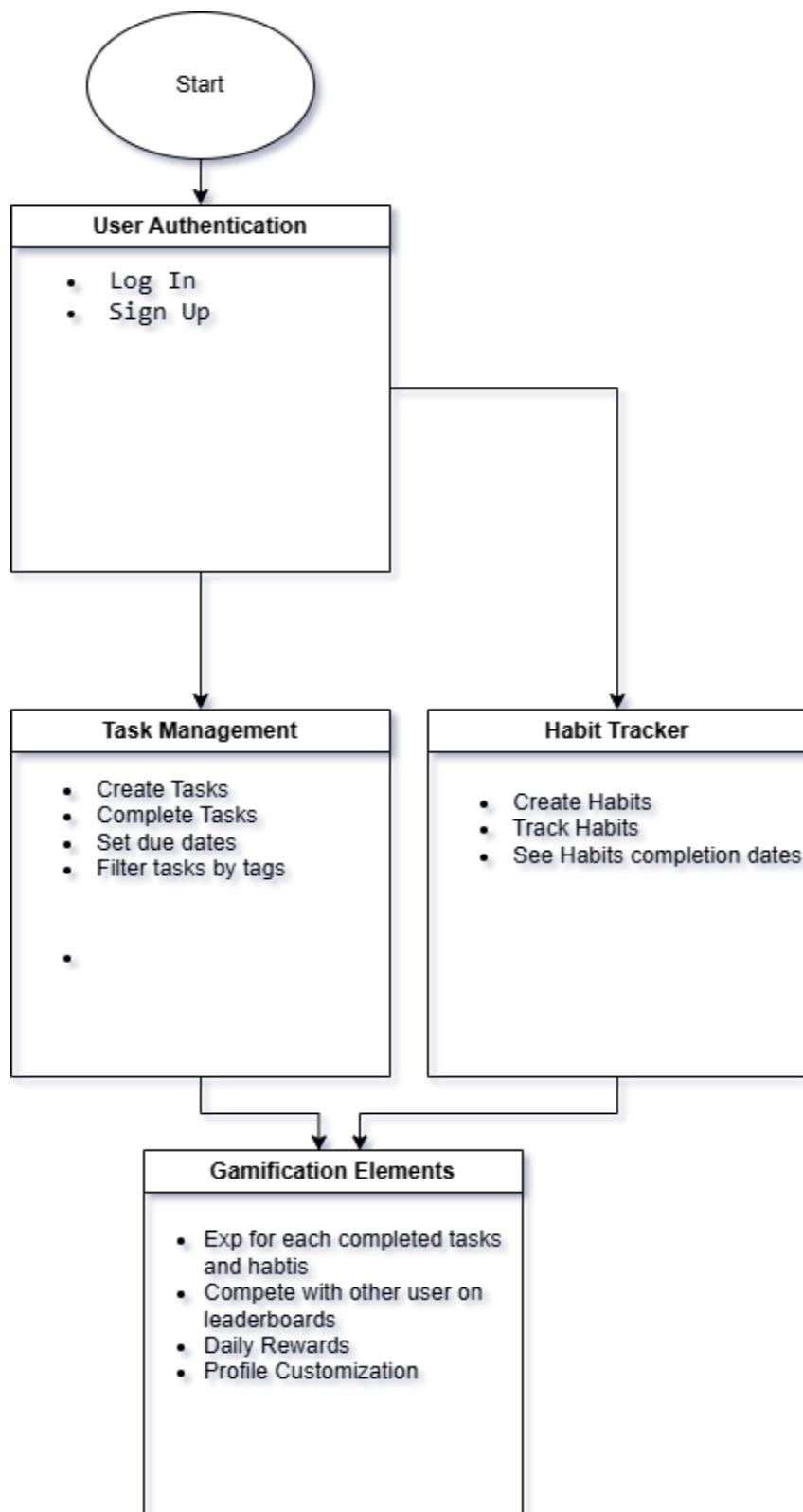


Figure 1: App Flowchart

3.2 Hardware and/or software components

Flutter

Flutter is a cross-platform framework by Google used for building natively compiled applications from a single codebase. It offers fast development with its "hot reload" feature and customizable widgets, making it ideal for creating the gamified to-do list's interactive and visually appealing interface.

Firebase

Firebase, also by Google, is a backend-as-a-service platform providing tools like authentication, real-time database, and analytics. It is used in this project to manage user authentication, store task and reward data, and ensure secure, scalable, and synchronized app performance.

3.3 Hardware and/or Software Implementation

The implementation of the gamified to-do list involved several stages, focusing on developing and integrating software modules using Flutter and Firebase.

1. **Frontend Development:**

The user interface was designed using Flutter, leveraging its customizable widgets to create a visually appealing and user-friendly experience. Features such as task lists, leaderboards, and reward systems were implemented to incorporate gamification elements effectively.

2. **Backend Integration:**

Firebase was used to handle authentication, real-time database management, and data synchronization across devices. The Firebase Realtime Database stores user tasks, progress, and achievements, ensuring that data is updated and accessible in real-time.

3. **Testing and Refinement:**

The application underwent iterative testing to identify and resolve bugs and ensure smooth interaction between the frontend and backend components.

Chapter 4 Investigation/Experiment, Result, Analysis and Discussion

The scope of this project primarily focused on the design, development, and initial implementation of the gamified to-do list application. Due to time and resource constraints, experiments to evaluate user engagement and performance metrics were not conducted during this phase.

Chapter 5 Impacts of the Project

5.1 Impact of this project on societal, health, safety, legal and cultural issues

The project promotes better time management and productivity, which can contribute to more organized personal and professional lives, ultimately benefiting society by fostering efficiency and responsibility.

5.2 Impact of this project on environment and sustainability

The app reduces dependency on physical task management tools, such as paper planners. Thus minimizing waste and contributing to environmental conservation.

The project promotes sustainable habits by helping users organize tasks efficiently, reducing time and energy waste. For example, planned task management can prevent unnecessary trips or resource usage, reducing carbon footprints.

Chapter 6 Project Management

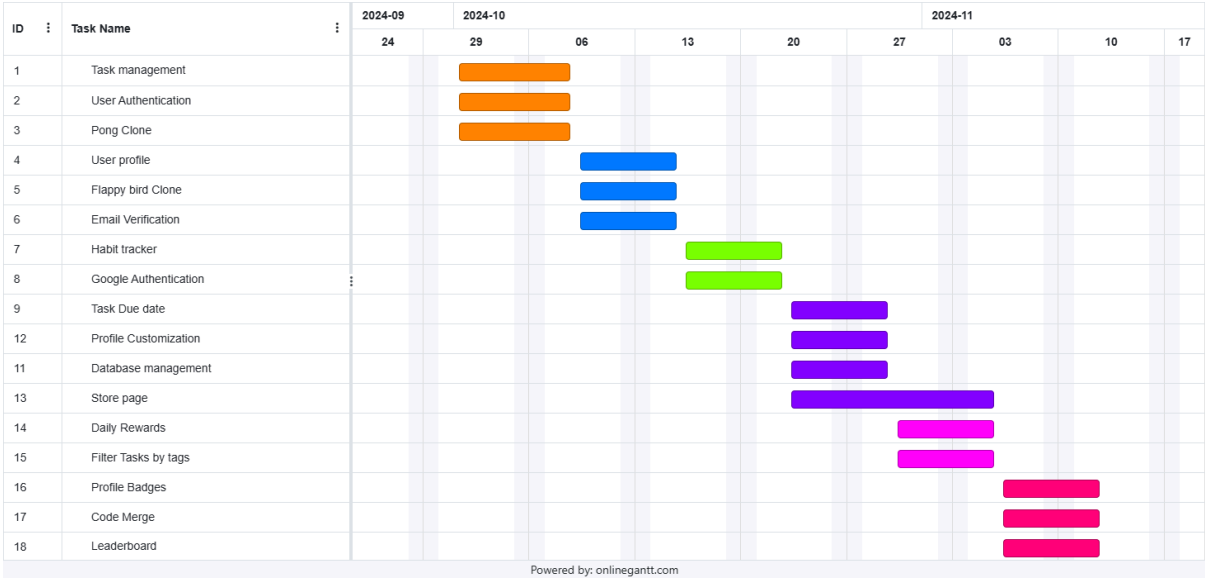


Figure 2 Gantt Chart

Chapter 7 Conclusion

7.1 Summary

The gamified to-do list projects gamified task management systems by adding elements like rewards, achievements and leaderboards. The goal of these features was to increase user engagement, user motivation as well as users' completion of the tasks. The application was built using Flutter as the front end and Firebase as the back end, thus allowing scalability and providing real-time updates. Although the project does an adequate job of showcasing the possibilities of gamification in enhancing productivity, more testing and refining of the project are needed in order to gauge its applicability and impact on users.

7.2 Limitations

The project has certain limitations: The project has certain limitations: As there were time constraints, extensive testing on a diverse group of users was not carried out which limits the understanding of how the app performs in real-life scenarios. The first version has only basic integration of gamification which might not satisfy all users. Metrics such as the rate of completion of tasks and the rate of returning customers have not been assessed quantitatively.

7.3 Future Improvement

The project has possibilities for further improvement that are quite considerable, especially with:

- Integrating more sophisticated task management capabilities such as subtasks, task notifications and priority.
- Adding analytics that would help users understand their productivity patterns and tendencies.
- Adapting the application for the web and the iOS platform would enhance its reach.

Hopefully, by overcoming these limitations and making the suggested improvements, the project should develop into a competent productivity application that can be easily embraced by a large number of users in the long run.